material in and out of the central flames. A deflagration tears through the star, creating a thermonuclear explosion so powerful that it rips the star apart, spectacularly and completely.

The explosion of a white dwarf star is so bright that it can briefly outshine its entire galaxy, and is visible to observers with telescopes billions of light-years away. Supernovae in distant parts of the Milky Way and nearby galaxies have even been seen without instruments, in ancient times, by the naked eye, *in the daytime*.[fn11]

It's a matter of some frustration in the astronomical community that, aside from this broad-brush-stroke picture, we still don't know *exactly* how Type Ia supernovae happen. There are ongoing debates about whether they're primarily caused by material falling onto the white dwarf from companion stars or by white dwarf collisions. Simulating the explosion ripping through the star is also computationally extremely difficult. Most simulations result in incredibly impressive visualizations of bubbling, churning stellar material without actually quite getting to the exploding part. But they're working on it. (Stars, it turns out, are complicated. Especially when quantum mechanics and thermonuclear explosions are both important.)

The thing that makes us think we can learn anything useful from Type Ia supernova observations is the fact that we can reasonably expect that white dwarfs[fn12] are pretty much always at the same mass when they go off. In 1930, a twenty-year-old prodigy physicist from India named Subrahmanyan Chandrasekhar was on a ship traveling to England to begin his studies at Cambridge when he casually revolutionized stellar evolution in his free time. By improving on existing calculations and adding in important effects from relativity, he discovered a hard limit on the mass of any star held up by electron degeneracy pressure. That limit, about 1.4 times the mass of the Sun, became known, appropriately, as the Chandrasekhar Limit. Any white dwarf that gains enough mass to exceed that limit is immediately doomed to explode spectacularly as a supernova. And now that we know that the physics of the explosion is always the same, we know how bright a Type Ia supernova is intrinsically, and can therefore figure out its distance.

When Chandrasekhar's ship finally reached the shore, his breakthrough tore through the scientific establishment like a detonation front of knowledge, forever changing our view of these weird and wonderful explosive stellar objects. (Though not everyone was convinced. Apparently,